Lowest common multiples and highest common factors

A Cambridge insert: prime factors, and the two numbers they build.

LESSON 38 | 1 × 40 MIN | MATEMATIKA 6, QO'SHIMCHA MAVZU | S7 1.3–1.4

LESSON CLOCK

10' | 12' | 12' | 6'

Prime factors	10 min
The HCF	12 min
The LCM	12 min
Where each is used	6 min

LEARNING OBJECTIVES

- Write a number as a product of prime factors.
- Find the HCF and the LCM of two numbers.
- Use the HCF to simplify a fraction and the LCM to add fractions.
- Recognise where each is needed in a word problem.

TEXTBOOK

National	pp. 99–102
Cambridge	Stage 7, pp. 18–24
Workbook	Exercise 1.3

01 Explanation

Prime factors

Every whole number above 1 is a product of primes, in exactly one way.

NUMBER	PRIME FACTORS	IN INDEX FORM
12	$2 \cdot 2 \cdot 3$	$2^2 \cdot 3$
18	$2 \cdot 3 \cdot 3$	$2 \cdot 3^2$
24	$2 \cdot 2 \cdot 2 \cdot 3$	$2^3 \cdot 3$
60	$2 \cdot 2 \cdot 3 \cdot 5$	$2^2 \cdot 3 \cdot 5$

DIVIDE BY THE SMALLEST PRIME THAT FITS, REPEATEDLY

$60 \div 2 = 30$, $30 \div 2 = 15$, $15 \div 3 = 5$, $5 \div 5 = 1$. The divisors used, in order, are the prime factorisation.

The HCF

The **highest common factor** is the product of the primes both numbers share, each to the lower of its two powers.

NUMBERS	FACTORISED	SHARED	HCF
$12, 18$	$2^2 \cdot 3, 2 \cdot 3^2$	$2 \cdot 3$	6
$24, 60$	$2^3 \cdot 3, 2^2 \cdot 3 \cdot 5$	$2^2 \cdot 3$	12
$8, 15$	$2^3, 3 \cdot 5$	nothing	1

AN HCF OF 1 MEANS THE NUMBERS ARE COPRIME

8 and 15 share no prime factor, so the fraction $\frac{8}{15}$ is already in its lowest terms.

The LCM

The **lowest common multiple** takes every prime that appears in either number, each to the higher of its powers.

NUMBERS	FACTORISED	TAKEN	LCM
$12, 18$	$2^2 \cdot 3, 2 \cdot 3^2$	$2^2 \cdot 3^2$	36
$24, 60$	$2^3 \cdot 3, 2^2 \cdot 3 \cdot 5$	$2^3 \cdot 3 \cdot 5$	120
$8, 15$	$2^3, 3 \cdot 5$	$2^3 \cdot 3 \cdot 5$	120

$$\text{HCF} \times \text{LCM} = \text{the product of the two numbers}$$

A CHECK THAT COSTS NOTHING

$6 \cdot 36 = 216 = 12 \cdot 18$ ✓. If the HCF and LCM you found do not multiply to the product of the two numbers, one of them is wrong.

Where each is used

PROBLEM	WHICH	WHY
simplifying $\frac{12}{18}$	HCF	divide both by 6
adding $\frac{1}{12} + \frac{1}{18}$	LCM	the denominator 36
the largest identical parcels from 24 pens and 60 pencils	HCF	12
two buses leaving every 12 and 18 minutes together again	LCM	36

> “CUTTING UP” NEEDS THE HCF; “MEETING AGAIN” NEEDS THE LCM
>
> The HCF is never bigger than either number; the LCM is never smaller. Comparing your answer with the two numbers tells you at once whether you found the right one.

02 Worked examples

EXAMPLE 1 **Find the HCF and LCM of 12 and 18.**

1. $12 = 2^2 \cdot 3$ and $18 = 2 \cdot 3^2$.
2. HCF: shared primes at the lower power — $2 \cdot 3 = 6$.
3. LCM: every prime at the higher power — $2^2 \cdot 3^2 = 36$.
4. Check: $6 \cdot 36 = 216 = 12 \cdot 18$ ✓

ANSWER

HCF 6, LCM 36

EXAMPLE 2

24 pens and 60 pencils are made into identical parcels with nothing left over. What is the largest number of parcels?

1. "Largest identical parcels" means the HCF.
2. $24 = 2^3 \cdot 3$, $60 = 2^2 \cdot 3 \cdot 5$.
3. HCF $= 2^2 \cdot 3 = 12$.
4. Twelve parcels, each with 2 pens and 5 pencils.

ANSWER

12 parcels

EXAMPLE 3

Two buses leave at $8{:}00$, one every 12 minutes and one every 18. When do they next leave together?

1. "Together again" means the LCM.
2. LCM of 12 and 18 is 36.
3. 36 minutes later, at $8{:}36$.

ANSWER

$8{:}36$

03 Interactive model

Ask for the HCF and LCM of 12 and 18 by listing factors and multiples first; the prime-factor method then arrives as a shortcut for something already understood.

Building the LCM from prime factors

DENOMINATOR	FACTORISED	EXTRA FACTOR NEEDED
12	$2^2 \cdot 3$	3
18	$2 \cdot 3^2$	2

LOWEST COMMON DENOMINATOR

$$2^2 \cdot 3^2 = 36$$

HCF $= 2 \cdot 3 = 6$, and $6 \cdot 36 = 12 \cdot 18$ ✓

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Factor	Bo‘luvchi	Делитель
Multiple	Karrali son	Кратное
Prime number	Tub son	Простое число
Prime factorisation	Tub ko‘paytuvchilarga ajratish	Разложение на простые множители
Highest common factor	EKUB	НОД
Lowest common multiple	EKUK	НОК
Common	Umumiy	Общий
Product	Ko‘paytma	Произведение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

12 in prime factors is:

- $2 \cdot 6$
- $2^2 \cdot 3$
- $3 \cdot 4$
- $2 \cdot 3^2$

The HCF of 12 and 18 is:

- 2
- 3
- 6
- 36

The LCM of 12 and 18 is:

- 6
- 36
- 54
- 216

The HCF of 8 and 15 is:

- 1
- 2
- 3
- 120

HCF × LCM equals:

- the sum
- the product of the numbers
- the difference
- nothing special

Two buses meeting again needs the:

- HCF
- LCM
- product
- difference

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. 12 in prime factors

   ANSWER $2^2 \cdot 3$

2. 18 in prime factors

   ANSWER $2 \cdot 3^2$

3. 24 in prime factors

   ANSWER $2^3 \cdot 3$

4. HCF of 12 and 18

   ANSWER 6

5. LCM of 12 and 18

   ANSWER 36

6. HCF of 8 and 15

   ANSWER 1

7. LCM of 8 and 15

   ANSWER 120

Medium · the standard question

7 PROBLEMS

1. 60 in prime factors

   ANSWER $2^2 \cdot 3 \cdot 5$

2. HCF of 24 and 60

   ANSWER 12

3. LCM of 24 and 60

   ANSWER 120

4. Simplify $\frac{12}{18}$ using the HCF

   ANSWER $\frac{2}{3}$

5. Add $\frac{1}{12} + \frac{1}{18}$ using the LCM

   ANSWER $\frac{5}{36}$

6. 24 pens and 60 pencils in identical parcels

   ANSWER 12 parcels

7. Buses every 12 and 18 minutes from $8{:}00$

   ANSWER $8{:}36$

Hard · stretch and reason

7 PROBLEMS

1. HCF and LCM of 36 and 48

   ANSWER 12 and 144

2. Check that HCF × LCM = $36 \cdot 48$

   ANSWER $12 \cdot 144 = 1728$ ✓

3. HCF of 105 and 154

   ANSWER 7

4. LCM of 4, 6 and 10

   ANSWER 60

5. The largest square tile that tiles a 60 by 84 cm floor exactly

   ANSWER 12 cm

6. Two lights flash every 15 and 25 seconds: together again after

   ANSWER 75 s

7. If the HCF of two numbers is one of them, what does that mean?

   ANSWER The smaller divides the larger

07 Homework

Homework — 5 tasks

Check every pair with HCF × LCM = the product of the numbers.

1. Write 45 and 75 as products of prime factors.
2. Find their HCF and LCM, and check with the product rule.
3. Simplify $\frac{45}{75}$ using the HCF.
4. Find the LCM of 6, 8 and 12.

5. Two runners lap a track in 40 and 60 seconds. After how long are they together at the start again?